

Burp Suite Web Application Security Home Lab Documentation

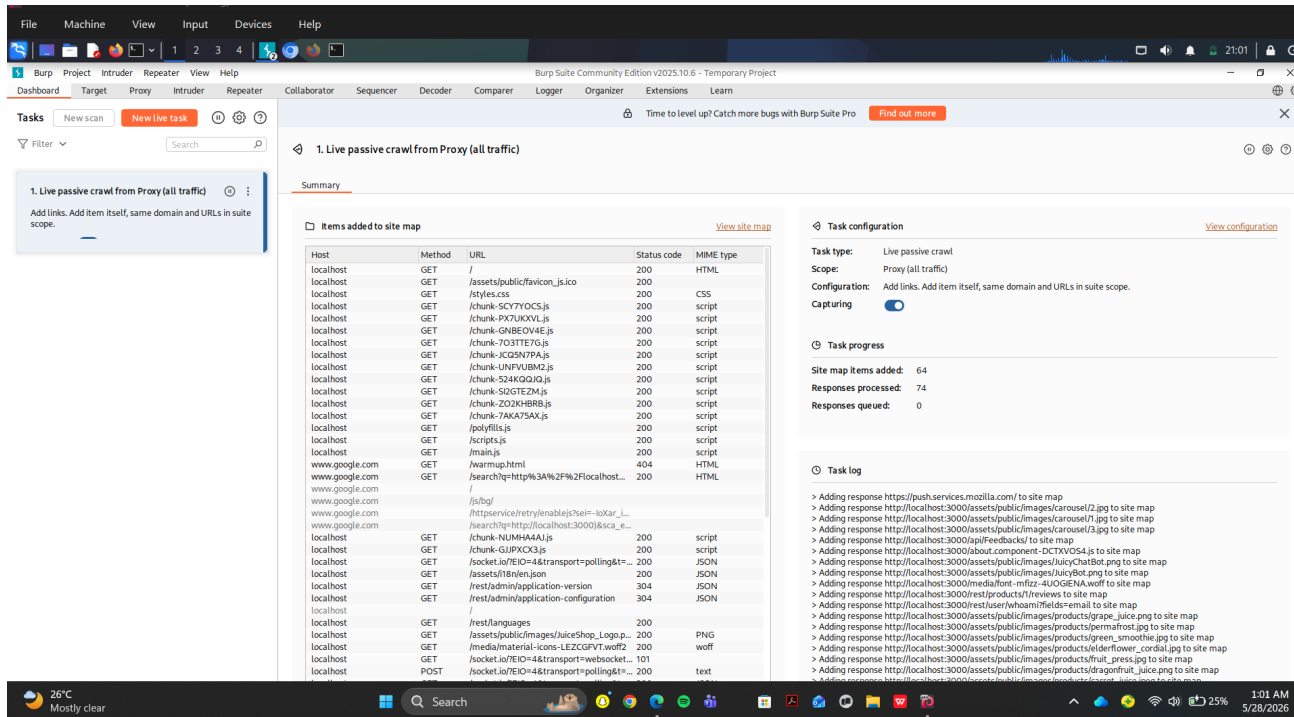
This document provides a professional overview of a completed cybersecurity home lab focused on web application security testing using Burp Suite Community Edition within a Kali Linux virtualized environment. The lab demonstrates practical exposure to proxy interception, passive crawling, WebSocket monitoring, HTTP request analysis, and Burp Intruder attack simulations.

1. Lab Environment

Component	Description
Operating System	Kali Linux 2025.4
Virtualization Platform	Oracle VirtualBox
Primary Tool	Burp Suite Community Edition v2025.10.6
Target Application	Localhost Web Application (Port 3000)
Testing Focus	Web Application Security Assessment

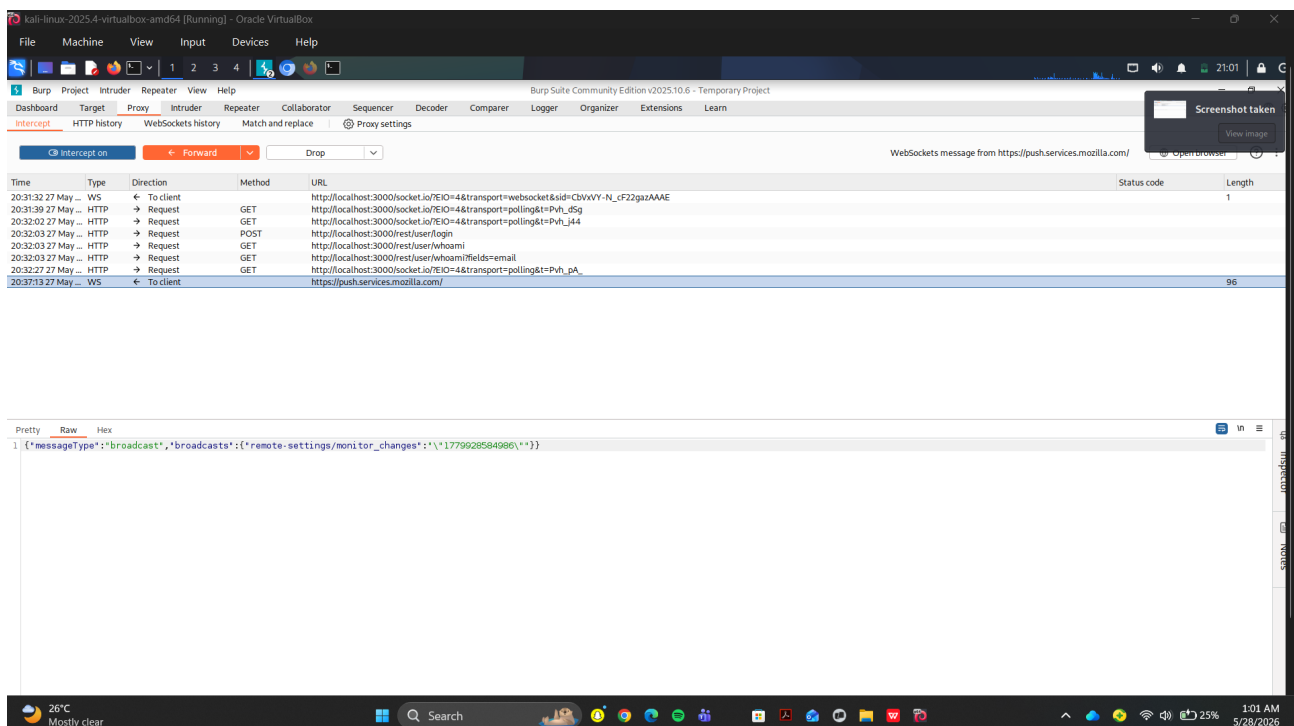
Passive Crawling and Site Mapping

This phase demonstrates the use of Burp Suite's live passive crawl functionality to automatically discover application resources and endpoints from proxied traffic. The scan identified multiple JavaScript files, images, API endpoints, authentication-related routes, and static assets. The site map expanded dynamically as traffic passed through the Burp proxy, allowing deeper visibility into the application's attack surface.



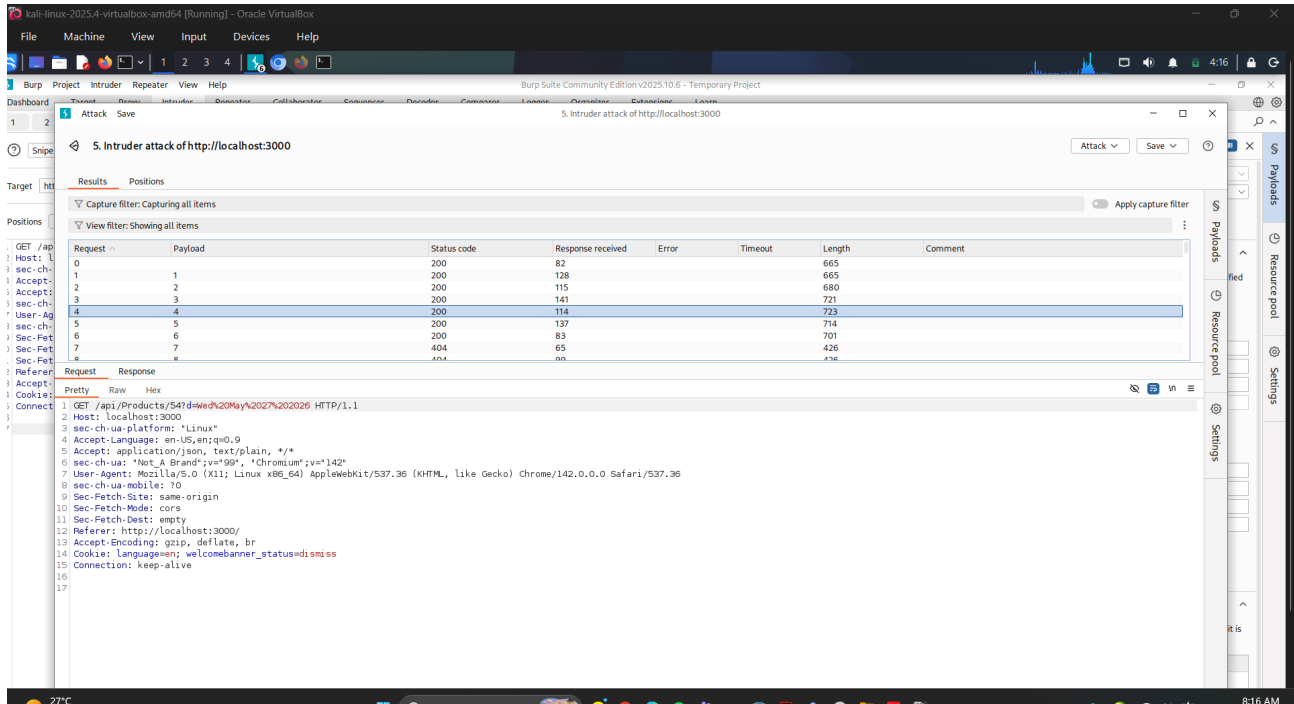
HTTP and WebSocket Interception

The proxy interception stage captured both HTTP requests and WebSocket traffic generated by the application. Requests such as login activity, socket communications, and API interactions were monitored. This provided insight into how the application handled real-time communication and session activity. WebSocket analysis is critical for identifying insecure message handling and authentication weaknesses.



Burp Intruder Attack Simulation

Burp Intruder was used to automate payload injection and parameter manipulation against the target application running on localhost:3000. Multiple payloads were tested against the endpoint, and the response length and status codes were analyzed to identify unusual behavior. This practical exercise demonstrates foundational skills in fuzzing, enumeration, and input validation testing.



2. Technical Skills Demonstrated

- Web application traffic interception and monitoring
- Passive reconnaissance and site mapping
- HTTP request and response analysis
- WebSocket communication inspection
- Intruder-based fuzzing and parameter testing
- Vulnerability assessment workflow exposure
- Use of Kali Linux for cybersecurity operations
- Virtualized lab environment management

3. Conclusion

This home lab provided hands-on experience with modern web application security assessment techniques using Burp Suite Community Edition. Through passive crawling, request interception, WebSocket monitoring, and Intruder attack simulations, practical exposure was gained in identifying application behavior, attack surfaces, and potential vulnerabilities. The exercise strengthened foundational penetration testing and web security analysis skills, forming

part of a growing cybersecurity portfolio.